

Chapter four

Analysis

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- During **requirements elicitation**, the client and developers define the purpose of the system. (Develop use cases)
 - During **analysis**, developers aim to produce a model of the system that is correct, complete, consistent, and unambiguous.
 - During **system design**, developers define the design goals of the project and decompose the system into smaller subsystems that can be realized by individual teams.

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- During **object design**, developers define solution domain objects to bridge the gap between the analysis model and the hardware/ software platform defined during system design.
 - During **implementation**, developers translate the solution domain model into source code.
 - During **testing**, developers find differences between the system and its models by executing the system with sample input data.



During requirements elicitation, the client and developers define the purpose of the system.

- The result of this phase is a Requirements Specification.
 - ◆ Written in natural language
- The Requirements Specification contains
 - ◆ Nonfunctional Requirements
 - ◆ Functional model
- In object oriented development, this tends to be represented by use cases and scenarios

Analysis Concepts(Entity, Boundary, and Control Objects)

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- The analysis object model consists of entity, boundary, and control objects.

Entity objects

- Represent the persistent information tracked by the system.
- **Definition:** Entity objects represent the core data and business logic of the application. They typically correspond to real-world concepts or entities that the system will manage.
- **Characteristics:**
 - They often have a persistent state (e.g., stored in a database).
 - They encapsulate business rules and validation logic.
 - Examples include classes like Customer, Order, or Product.

Boundary objects

- represent the interactions between the actors and the system.
- **Definition:** Boundary objects act as the interface between the system and the external environment, which can include users or other systems. They handle communication and user interaction.
- **Characteristics:**
- They manage input and output operations, such as displaying data to users or collecting user input.
- They often involve user interface components, such as forms and dialogs.
- Examples include classes like `UserInterface`, `WebController`, or `APIEndpoint`.

Control objects

- Represent the tasks that are performed by the user and supported by the system.
- **Definition:** Control objects manage the flow of the application and coordinate the interactions between entity and boundary objects. They encapsulate the application's logic and control its behavior.
- **Characteristics:**
 - They often implement the use cases or application workflows.
 - They may handle the orchestration of multiple entity objects to fulfill a particular function.
 - Examples include classes like `OrderController`, `PaymentProcessor`, or `SessionManager`.

Analysis Concepts(Association Multiplicity Revisited)

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- Modeling with UML the end of an association can be labeled by a set of integers called **multiplicity**.
- The multiplicity indicates the number of links that can legitimately originate from an instance of the class connected to the association end.

There are three types of associations:

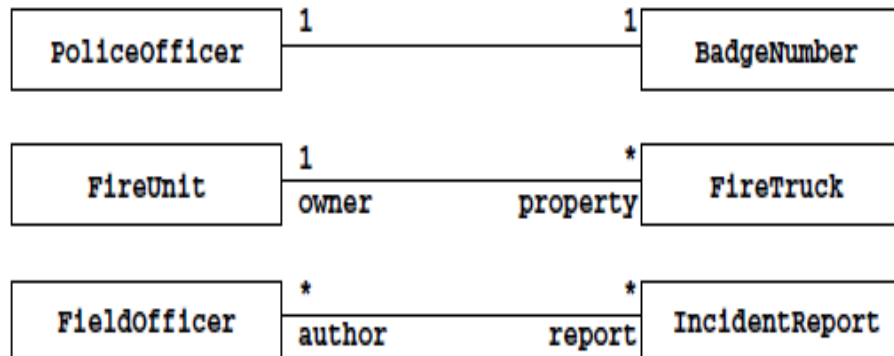
1. A **one-to-one Association**: has a multiplicity 1 on each end.

- ✓ A one-to-one association between two classes means that exactly one link exists between instances of each class.

Analysis Concepts(Association Multiplicity Revisited)

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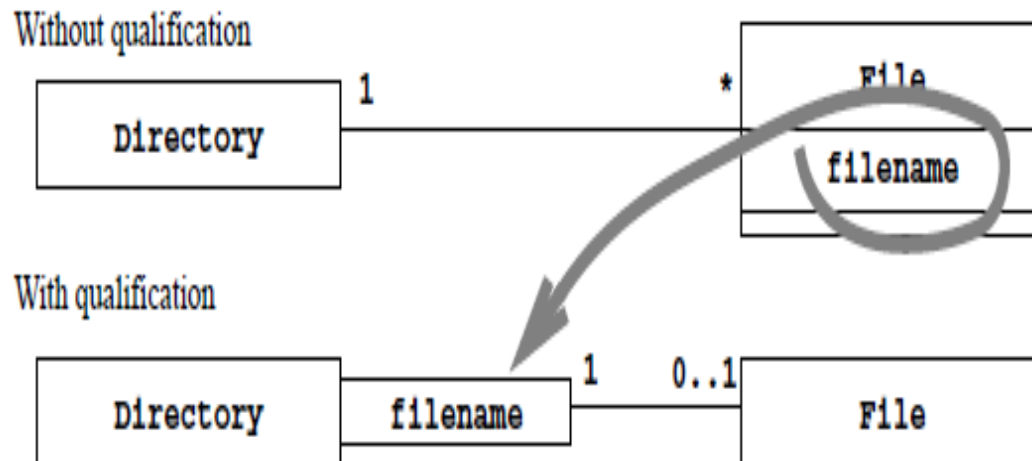
- 2.A **one-to-many association**: has a multiplicity 1 on one end and 0..n (also represented by a star) or 1..n on the other.
- 3.A **many-to-many association** has a multiplicity 0..n or 1..n on both ends.



Analysis Concepts(Qualified Associations: Reducing Multiplicity)

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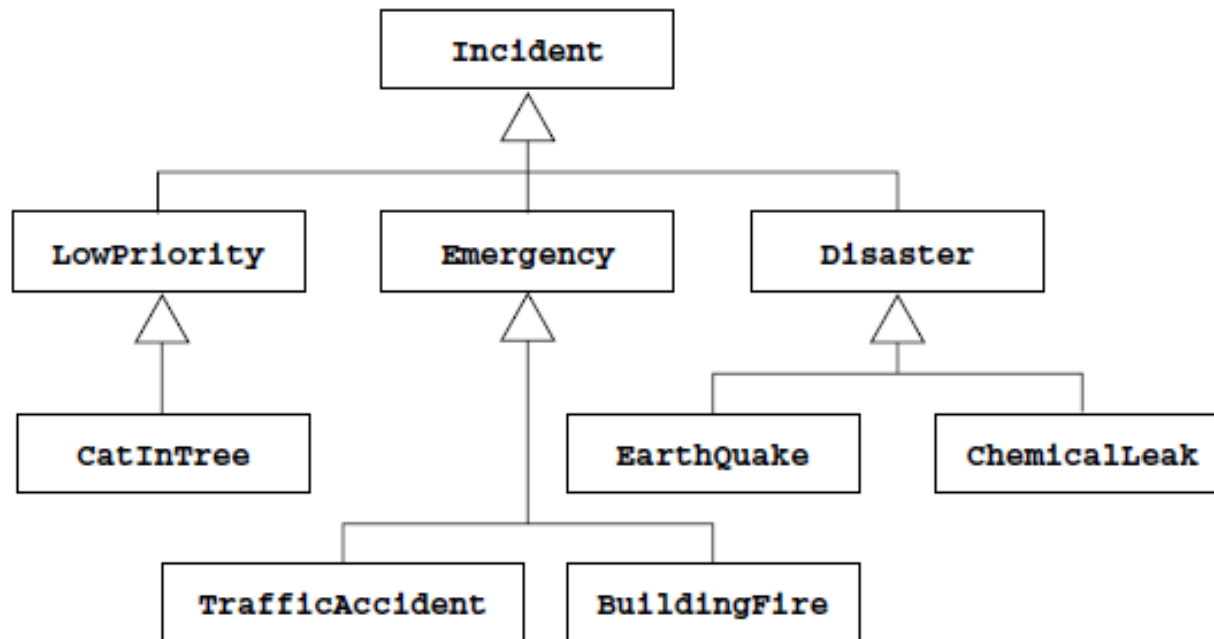
- **Qualification** is a technique for **reducing** multiplicity by using **keys**.
- Associations with a 0..1 or 1 multiplicity are easier to understand than associations with a 0..n or 1..n multiplicity.
- Often, in the case of a one-to-many association, objects on the “many” side can be distinguished from one another using a name.
 - ▣ E.g



Analysis Concepts(Generalization)

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- **Generalization** enables to organize concepts into hierarchies.
- At the top of the hierarchy is a general concept and at the bottom of the hierarchy are the most specialized concepts.
- There may be any number of intermediate levels in between, covering more-or-less generalized concepts.



Analysis Activities: From Use Cases to Objects

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The activities that transform the use cases and scenarios produced during requirements elicitation into an analysis model.

- Identifying entity objects
- Identifying boundary objects
- Identifying control objects
- Mapping use cases to objects
- Identifying associations among objects
- Identifying object attributes
- Modeling generalization relationships
- Reviewing the analysis model

Analysis Activities: Identifying Entity Objects

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- During Requirements Elicitation, participating objects are found by examining each use case and identifying candidate objects.
- Natural language analysis [Abbott, 1983] is an intuitive set of heuristics for identifying objects, attributes, and associations from a system specification.

Table 5-1 Abbott's heuristics for mapping parts of speech to model components [Abbott, 1983]

Part of speech	Model component	Examples
Proper noun	Object	Alice
Common noun	Class	FieldOfficer
Doing verb	Operation	Creates, submits, selects
Being verb	Inheritance	Is a kind of, is one of either
Having verb	Aggregation	Has, consists of, includes
Modal verb	Constraints	Must be
Adjective	Attribute	Incident description

Analysis Activities: Identifying Boundary Objects

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- Boundary objects represent the system interface with the actors.
- In each use case, each actor interacts with at least one boundary object.
- The boundary object collects the information from the actor and translates it into an interface neutral form that can be used by the entity objects and also by the control objects.

Analysis Activities: Identifying Control Objects

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- Control objects are responsible for **coordinating** boundary and entity objects.
- Control objects usually do not have a concrete counterpart in the real world.
- There is often a close relationship between a use case and a control object.
- Control objects is responsible for collecting information from the **boundary** objects and dispatching it to **entity** objects

Analysis Activities: Interactions Between Objects: Sequence Diagrams

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- A sequence diagram ties use cases with objects.
- It shows how the behavior of a use case (or scenario) is distributed among its participating objects.
- They represent, however, another shift in perspective and allow the developers to find missing objects or grey areas in the system specification.

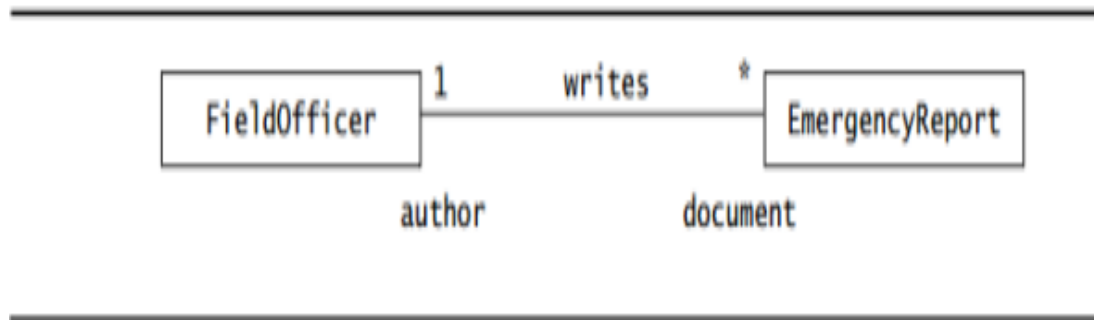
Identifying Associations

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- An association shows a relationship between two or more classes.

Associations have several properties:

- Name: to describe the association between the two classes.
- Association names are optional and need not be unique globally.
- Role at each end, identifying the function of each class with respect to the associations (e.g., author is the role played by FieldOfficer in the Writes association).
- A multiplicity at each end, identifying the possible number of instances



Identifying Attributes

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- Attributes are **properties** of individual objects.

Attributes have:

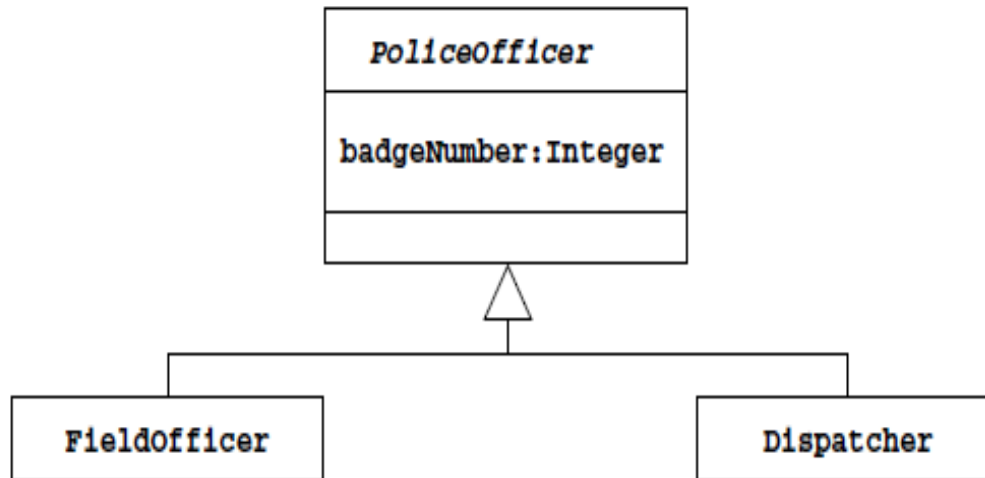
- A name identifying them within an object
- A brief description.
- A type describing the legal values it can take.

EmergencyReport
emergencyType:{fire,traffic,other} location:String description:String

Analysis Activities: Modeling Generalization Relationships Between Objects

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- Generalization is used to **eliminate** redundancy from the analysis model.
- If two or more classes share attributes or behavior, the similarities are consolidated into a **superclass**.



Reviewing the Analysis Model

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- The analysis model is built incrementally and iteratively.
- The analysis model is seldom correct or even complete on the first pass.
- Several iterations with the **client** and the **user** are necessary before the analysis model converges toward a correct specification usable by the **developers** for design and implementation.

Documenting Analysis

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- The RAD(Requirement Analysis Documenting), once completed and published, will be baseline and put under configuration management.
- The revision history section of the RAD will provide a history of changes including the author responsible for each change, the date of the change, and brief description of the change.

Understanding Document Analysis

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- Document analysis is a systematic and essential process for assessing printed and electronic documents, from conventional paper-based materials to digital files sent over the Internet.
- It is a part of modern business processes and a fundamental technique used in research, business, and data analysis to extract valuable information and insights from various document forms.

Document analysis in various contexts

□ Research

- In academia and scientific research, document analysis is essential for examining historical records, textual data, and archival documents to extract relevant information for scholarly investigations.
- It aids researchers in understanding societal trends, cultural shifts, and historical events by analyzing primary sources according to the research proposal.

□ Business

- Document analysis is crucial in contract review, financial analysis, market research, and competitive intelligence in the corporate world.
- Scrutinizing business documents such as contracts, financial reports, and market studies can help organizations make informed decisions, identify risks, and capitalize on opportunities.

- Analysts can better understand consumer sentiment, market trends, and competitor strategies by analyzing unstructured data from customer feedback, social media posts, and surveys.
- Document analysis is a versatile and indispensable method for extracting valuable information and insights from various documents.
- It contributes to informed decision-making, scholarly research, and data-driven strategies in multiple domains.